

CHAPTER 1: MULTIVARIATE NORMAL DISTRIBUTION

Q1) Theorem 2

Statements (a) - (d) are equivalent.

X is a random vector, $a \in \mathbb{R}^{n+1}$, $\Sigma \in \mathbb{R}^{n \times n}$, $n, n.d$
(non-negative definite)

$$(a) X \sim N(\mu, \Sigma) \text{ i.e., } X = \Sigma^{1/2} Z + \mu, Z \sim N(0, I_n)$$

$$(b) X \stackrel{d}{=} AZ + \mu, Z \sim N(0, I_K), A \in \mathbb{R}^{n \times K}, AA^T = \Sigma$$

$$(c) \text{ The mgf is given by: } M_X(t) = \exp\left\{\mu^T t + \frac{1}{2} t^T \Sigma t\right\} \quad (t \in \mathbb{R})$$

$$(d) a^T X \sim N(a^T \mu, a^T \Sigma a) \quad \forall a \in \mathbb{R}^n$$

(proof)

$$(c) \text{ Let } Z \sim N(0, I_K)$$

$$\begin{aligned} M_Z(t) &= E[e^{t^T Z}] = E[e^{t_1 z_1 + \dots + t_K z_K}] = E[e^{t_1 z_1}] \dots E[e^{t_K z_K}] \\ &= \exp\left\{-\frac{1}{2}(t_1^2 + \dots + t_K^2)\right\} \\ &= \exp\left\{-\frac{1}{2} t^T t\right\} \end{aligned}$$

$$X = \Sigma^{1/2} Z + \mu \quad \frac{dz}{dx} = \Sigma^{-1/2}, \left| \frac{dz}{dx} \right| = \det(\Sigma^{-1/2})$$

$$\begin{aligned} \therefore M_X(t) &= E[\exp(t^T X)] = E[\exp(t^T (\Sigma^{1/2} Z + \mu))] \\ &= E[\exp(t^T \Sigma^{1/2} Z + t^T \mu)] \\ &= \exp\{t^T \mu\} \exp\left\{(\Sigma^{1/2} t)^T (\Sigma^{1/2} Z)\right\} \\ &= \exp\left\{\mu^T t + \frac{1}{2} t^T \Sigma t\right\} \end{aligned}$$

$$\therefore (c) \Leftrightarrow (a)$$

$$\text{Let } X = AZ + \mu$$

$$\begin{aligned} M_X(t) &= E[\exp\{(\lambda Z + \mu)^T t\}] = e^{\mu^T t} E[\exp\{Z^T A^T t\}] = e^{\mu^T t} E[\exp\{Z^T (A^T t)\}] \\ \exp\left\{\mu^T t + \frac{1}{2} t^T A A^T t\right\} &= \exp\left\{\mu^T t + \frac{1}{2} t^T \Sigma t\right\} \\ \therefore A A^T &= \Sigma \end{aligned}$$

$$\therefore (c) \Leftrightarrow (b)$$

$$\begin{aligned} M_{a^T X}(t) &= E[\exp\{a^T X\} t] = E[\exp\{a^T (AZ + \mu)\} t] \\ &= \exp\left\{\mu^T (a^T t) + \frac{1}{2} (a^T t)^T \Sigma (a^T t)\right\} \\ &= \exp\left\{(a^T \mu)^T t + \frac{1}{2} t^T a^T \Sigma a t\right\} \end{aligned}$$

$$\therefore E[a^T X] = a^T \mu, \text{Var}(a^T X) = a^T \Sigma a$$

$$\therefore a^T X \sim N(a^T \mu, a^T \Sigma a)$$

$$\therefore (c) \Leftrightarrow (d)$$

Q2) Prove Theorem 2

$X \sim N_n(\mu, \Sigma)$, Σ : Symmetric and positive definite matrix.

$$\text{pdf of } X: f_X(x) = |\det(2\pi\Sigma)|^{-1/2} \exp\left\{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right\}, x \in \mathbb{R}^n$$

$$f_Z(z) = (2\pi)^{-n/2} \exp\left\{-\frac{1}{2} z^T z\right\} = \det(2\pi I_n)^{-1/2} \exp\left\{-\frac{1}{2}(z_1^2 + \dots + z_n^2)\right\}$$

$$X = \Sigma^{1/2} Z + \mu \quad \frac{dz}{dx} = \Sigma^{-1/2}, \left| \frac{dz}{dx} \right| = |\det(\Sigma^{-1/2})|$$

$$\begin{aligned} \therefore f_X(x) &= \det(2\pi I_n)^{-1/2} \exp\left\{-\frac{1}{2}(\Sigma^{-1/2})^T (x-\mu)\right\} \left[(\Sigma^{-1/2})^T (x-\mu)\right]^T |J_U| \\ &= |\det(2\pi I_n)|^{-1/2} \det(\Sigma^{-1/2}) \exp\left\{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right\} \\ &= |\det(2\pi\Sigma)|^{-1/2} \exp\left\{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right\} \end{aligned}$$

Q3) Prove Theorem 4

$$X \sim N(\mu, \Sigma)$$

$$(a) E[X] = \mu, \text{Var}(X) = \Sigma$$

$$M_Z(t) = \exp\left\{\frac{1}{2} t^T t\right\} = \exp\left\{\frac{1}{2}(t_1^2 + \dots + t_n^2)\right\}$$

$$C_Z(t) = \frac{1}{2}(t_1^2 + \dots + t_n^2)$$

$$C_Z^1(t) = [t_1 \dots t_n]^T$$

$$C_Z^n(t) = I_n$$

$$\therefore E[Z] = 0 \text{ and } \text{Var}(Z) = I_n$$

$$E[X] = E[\Sigma^{1/2} Z + \mu] = \Sigma^{1/2} E[Z] + \mu = \mu$$

$$\text{Var}(X) = \text{Var}(\Sigma^{1/2} Z + \mu) = \Sigma^{1/2} \text{Var}(Z) (\Sigma^{1/2})^T = \Sigma^{1/2} \Sigma^{1/2} = \Sigma$$

$$(b) \forall A \in \mathbb{R}^{m \times n} \text{ and } b \in \mathbb{R}^m: AX + b \sim N_n(A\mu + b, A\Sigma A^T)$$

$$\text{We know that } X = \Sigma^{1/2} Z + \mu: AX + b = A(\Sigma^{1/2} Z + \mu) + b = (\underbrace{A\Sigma^{1/2}}_{(\Sigma^{1/2})^T}) Z + \underbrace{A\mu + b}_{\mu'}$$

$$\begin{aligned} X' = AX + b &= (\Sigma^{1/2})^T Z + \mu' \sim N(\mu', (\Sigma^{1/2})^T \Sigma^{1/2}) \\ &\stackrel{d}{=} N(\mu', A\Sigma^{1/2} \Sigma^{1/2} A^T) \stackrel{d}{=} N(\mu', A\Sigma A^T) \end{aligned}$$

$$(c) \text{ if } X_1 \sim N_n(\mu_1, \Sigma_1), X_2 \sim N_m(\mu_2, \Sigma_2)$$

• Approach #1

$$E\left[\begin{pmatrix} X_1 \\ X_2 \end{pmatrix}\right] = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \text{Var}\left(\begin{pmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) \end{pmatrix}\right) = \begin{pmatrix} \Sigma_1 & 0 \\ 0 & \Sigma_2 \end{pmatrix}$$

• Approach #2

$$\therefore X_1 \text{ and } X_2 \text{ are independent, } M_{X_1, X_2}(t_1, t_2) = M_{X_1}(t_1) \cdot M_{X_2}(t_2)$$

$$\begin{aligned} M_{X_1}(t_1) M_{X_2}(t_2) &= \exp\left\{\mu_1^T t_1 + \frac{1}{2} t_1^T \Sigma_1 t_1\right\} \exp\left\{\mu_2^T t_2 + \frac{1}{2} t_2^T \Sigma_2 t_2\right\} \\ &= \exp\left\{\begin{bmatrix} \mu_1 & \mu_2 \end{bmatrix} \begin{bmatrix} t_1 \\ t_2 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} t_1 & t_2 \end{bmatrix} \begin{bmatrix} \Sigma_1 & 0 \\ 0 & \Sigma_2 \end{bmatrix} \begin{bmatrix} t_1 \\ t_2 \end{bmatrix}\right\} \end{aligned}$$

$$\therefore \mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \Sigma = \begin{bmatrix} \Sigma_1 & 0 \\ 0 & \Sigma_2 \end{bmatrix}$$

Q4) Prove Theorem 5

(a) We decompose the random vector $X \sim N_n(\mu, \Sigma)$ so that

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim N \left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \right)$$

X_1 and X_2 are independent if and only if $\text{Cov}(X_1, X_2) = \Sigma_{12} = \Sigma_{21}^T = 0$

In this case, we get to know that $X_1 \sim N_1(\mu_1, \Sigma_{11})$ and $X_2 \sim N(\mu_2, \Sigma_{22})$

(b) $A_1 \in \mathbb{R}^{m_1 \times n}$, $A_2 \in \mathbb{R}^{m_2 \times n}$

$A_1 X$ and $A_2 X$ are independent iff $\text{Cov}(A_1 X, A_2 X) = A_1 \Sigma A_2^T = 0$

(proof)

$$(a) M_{X_1, X_2}(t_1, t_2) = \exp \left\{ \mu_1^T t_1 + \mu_2^T t_2 + \frac{1}{2} (t_1^T \Sigma_{11} t_1 + t_2^T \Sigma_{22} t_2 + t_1^T \Sigma_{12} t_2 + t_2^T \Sigma_{21} t_1) \right\}$$

$$M_{X_1, X_2}(t_1, t_2 = 0) = \exp \left\{ \mu_1^T t_1 + \frac{1}{2} t_1^T \Sigma_{11} t_1 \right\} = M_{X_1}(t_1)$$

$$M_{X_1, X_2}(t_1 = 0, t_2) = \exp \left\{ \mu_2^T t_2 + \frac{1}{2} t_2^T \Sigma_{22} t_2 \right\} = M_{X_2}(t_2)$$

$\therefore$ If X_1 and X_2 are independent $M_{X_1, X_2}(t_1, t_2) = M_{X_1}(t_1) \cdot M_{X_2}(t_2) \Leftrightarrow \Sigma_{12} = \Sigma_{21}^T = 0$

$$(b) \begin{pmatrix} A_1 X_1 \\ A_2 X_2 \end{pmatrix} = \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = AX$$

$$\begin{aligned} \therefore AX &\sim N(A\mu, A\Sigma A^T) \stackrel{d}{=} N \left(\begin{pmatrix} A_1 \mu_1 \\ A_2 \mu_2 \end{pmatrix}, \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix} \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \begin{pmatrix} A_1^T & 0 \\ 0 & A_2^T \end{pmatrix} \right) \\ &\stackrel{d}{=} N \left(\begin{pmatrix} A_1 \mu_1 \\ A_2 \mu_2 \end{pmatrix}, \begin{pmatrix} A_1 \Sigma_{11} A_1^T & A_1 \Sigma_{12} A_2^T \\ A_2 \Sigma_{21} A_1^T & A_2 \Sigma_{22} A_2^T \end{pmatrix} \right) \end{aligned}$$

$\therefore A_1 X_1$ and $A_2 X_2$ are independent iff $A_1 \Sigma_{12} A_2^T = (A_2 \Sigma_{21} A_1^T)^T = 0$

Q5) Prove Theorem 6

Suppose, $\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim N \left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \right)$, Σ_{22} is positive definite

$$(a) \Sigma_{11.2} = \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}$$

The conditional distribution of X_1 under $X_2 = x_2$ is given by:

$$(X_1 | X_2 = x_2) \sim N(\mu_1 + \Sigma_{12} \Sigma_{22}^{-1} (x_2 - \mu_2), \Sigma_{11.2})$$

(b) The Marginal distribution of X_2 is given as $X_2 \sim N(\mu_2, \Sigma_{22})$

$$\text{Let } A = \begin{pmatrix} I_n & -\Sigma_{12} \Sigma_{22}^{-1} \\ 0 & I_n \end{pmatrix}, AX = \begin{pmatrix} I_n & -\Sigma_{12} \Sigma_{22}^{-1} \\ 0 & I_n \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = \begin{pmatrix} X_1 - \Sigma_{12} \Sigma_{22}^{-1} X_2 \\ X_2 \end{pmatrix} = \begin{pmatrix} Y \\ X_2 \end{pmatrix}$$

$$Y \sim N(A\mu, A\Sigma A^T) \stackrel{d}{=} N \left(\begin{pmatrix} \mu_1 - \Sigma_{12} \Sigma_{22}^{-1} \mu_2 \\ \mu_2 \end{pmatrix}, \underbrace{\begin{pmatrix} I_n & -\Sigma_{12} \Sigma_{22}^{-1} \\ 0 & I_n \end{pmatrix} \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \begin{pmatrix} I_n & 0 \\ -\Sigma_{22}^{-1} \Sigma_{21} & I_n \end{pmatrix}}_{(*)} \right)$$

$$* \begin{bmatrix} I_n & -\Sigma_{12} \Sigma_{22}^{-1} \\ 0 & I_n \end{bmatrix} \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} \begin{bmatrix} I_n & 0 \\ -\Sigma_{22}^{-1} \Sigma_{21} & I_n \end{bmatrix}$$

$$= \begin{bmatrix} \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} & 0 \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} \begin{bmatrix} I_n & 0 \\ -\Sigma_{22}^{-1} \Sigma_{21} & I_n \end{bmatrix}$$

$$= \begin{bmatrix} \Sigma_{11.2} & 0 \\ 0 & \Sigma_{22} \end{bmatrix} \therefore X_2 \text{ and } Y \text{ are independent.}$$

$$+ \text{Var}(Y) = \Sigma_{11.2}$$

$$\begin{aligned} (X_1 | X_2 = x_2) &= \{Y + \Sigma_{12} \Sigma_{22}^{-1} X_2 | X_2 = x_2\} \\ &= Y + \Sigma_{12} \Sigma_{22}^{-1} x_2 \\ &\sim N(E[Y] + \Sigma_{12} \Sigma_{22}^{-1} x_2, \text{Var}(Y)) \\ &\stackrel{d}{=} N(\mu_1 - \Sigma_{12} \Sigma_{22}^{-1} \mu_2 + \Sigma_{12} \Sigma_{22}^{-1} x_2, \Sigma_{11.2}) \\ &\stackrel{d}{=} N(\mu_1 + \Sigma_{12} \Sigma_{22}^{-1} (x_2 - \mu_2), \Sigma_{11.2}) \end{aligned}$$

Q6) Prove Theorem 8

$$E[X^T A X] = \text{tr}(A \Sigma) + \mu^T A \mu$$

$$\begin{aligned} (\text{proof}) \quad E[X^T A X] &= E[\text{tr}(X^T A X)] = E[\text{tr}(A X X^T)] \\ &= \text{tr}(E[A X X^T]) \\ &= \text{tr}(A E[X X^T]) \\ &= \text{tr}(A \{ \text{Var}(X) - E[X] E[X]^T \}) \\ &= \text{tr}(A \Sigma - A \mu \mu^T) \\ &= \text{tr}(A \Sigma) - \text{tr}(A \mu \mu^T) \\ &= \text{tr}(A \Sigma) - \text{tr}(\mu^T A \mu) \\ &= \text{tr}(A \Sigma) + \mu^T A \mu \quad \square \end{aligned}$$

Q7) Prove Example 9

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{1}{n-1} X^T (I_n - \Pi_n) X, \quad X_i \stackrel{iid}{\sim} (\mu_i, \sigma^2) \quad \mu_i = \mu_j \quad \forall i, j$$

$$E[S^2] = \frac{1}{n-1} E[X^T (I_n - \Pi_n) X] = \frac{1}{n-1} \{ \text{tr}((I_n - \Pi_n) \Sigma) + \mu^T (I_n - \Pi_n) \mu \}$$

$$\begin{aligned} * \Sigma &= (\text{Cov}(X_i, X_j))_{\substack{i, j \\ 1 \leq i \leq n \\ 1 \leq j \leq n}} = \sigma^2 I_n & \text{tr}(I_n - \Pi_n) &= \text{rank}(I_n - \Pi_n) = n-1 \\ & \text{tr}(\Sigma) &= n \sigma^2 \\ & \text{tr}((I_n - \Pi_n) \Sigma) &= \text{tr}(\sigma^2 (I_n - \Pi_n)) \\ & &= \sigma^2 \text{tr}(I_n - \Pi_n) = \sigma^2 (n-1) \end{aligned}$$

$$\begin{aligned} \mu^T (I_n - \Pi_n) \mu &= [(I_n - \Pi_n) \mu]^T [(I_n - \Pi_n) \mu] \\ &= [\mu - \bar{\mu}]^T [\mu - \bar{\mu}] = 0 \end{aligned}$$

$$\therefore E[S^2] = \frac{1}{n-1} \sigma^2 (n-1) = \sigma^2$$

Q8) Prove Theorem 10

$$X \sim N_n(\mu, \Sigma), A^T = A$$

$$\text{Cov}(X, X^T A X) = 2 \Sigma A \mu$$

$$x = \Sigma^{1/2} Z + \mu$$

$$\text{Cov}(\Sigma^{1/2} Z + \mu, (\Sigma^{1/2} Z + \mu)^T A (\Sigma^{1/2} Z + \mu))$$

$$= \text{Cov}(\Sigma^{1/2} Z, Z^T \Sigma^{1/2} A \Sigma^{1/2} Z) + \text{Cov}(\Sigma^{1/2} Z, \mu^T A \Sigma^{1/2} Z) \times 2$$

$$= \Sigma^T \text{Cov}(Z, Z^T A^* Z) + \Sigma^{1/2} \text{Cov}(Z, \Sigma^{1/2} A^T \mu) \times 2$$

$$= 0 + 2 \Sigma A \mu$$

$$\text{let } A^* = (a_{ij})_{\substack{1 \leq i \leq n \\ 1 \leq j \leq n}}. \text{Cov}(Z, Z^T A^* Z) = (\text{Cov}(z_i, a_{jk} z_j z_k))_{1 \leq i, j, k \leq n}$$

$$\text{Case ① } \text{Cov}(z_i, a_{jk} z_j z_k) \quad (i \neq j, i \neq k, j \neq k)$$

$$= a_{jk} \text{Cov}(z_i, z_j z_k)$$

$$= a_{jk} [E[z_i z_j z_k] - E[z_i] E[z_j z_k]]$$

$$= a_{jk} [E[z_i] E[z_j] E[z_k] - E[z_i] E[z_j] E[z_k]] = 0$$

$$\text{Case ② } \text{Cov}(z_i, a_{ij} z_j z_k)$$

$$= a_{ij} \text{Cov}(z_i, z_i z_k) = a_{ij} [E[z_i^2 z_k] - E[z_i] E[z_i z_k]]$$

$$= a_{ij} [E[z_i^2] E[z_k] - E[z_i]^2 E[z_k]] = 0$$

$$\text{Case ③ } \text{Cov}(z_i, a_{ii} z_i z_i)$$

$$= a_{ii} \text{Cov}(z_i, z_i^2) = a_{ii} [E[z_i^3] - E[z_i] E[z_i^2]] = 0$$

$$\therefore \text{Cov}(Z, Z^T A Z) = 0$$

Q9) Prove Theorem 11

$$X \sim N_n(\mu, \Sigma)$$

$$(a) A \in \mathbb{R}^{m \times n}, B \in \mathbb{R}^{n \times n}, B^T = B \Rightarrow A \Sigma B = 0 \Leftrightarrow AX \text{ and } X^T B X \text{ are independent}$$

$$(b) A, B \in \mathbb{R}^{n \times n}, A^T = A, B^T = B \Rightarrow A \Sigma B = 0 \Leftrightarrow X^T A X \text{ and } X^T B X \text{ are independent.}$$

(proof)

$$(a) B^T = B \quad \therefore B = P \Lambda P^T, \Lambda = \text{diag}(\lambda_1, \dots, \lambda_n), P P^T = P^T P = I_n$$

($\Rightarrow$)

$$\text{Cov}(AX, BX) = A \Sigma B^T = A \Sigma B = 0 \quad \therefore AX \text{ and } BX \text{ are mutually independent}$$

$$A \Sigma B^T = A \Sigma (P \Lambda P^T)^T = A \Sigma P \Lambda P^T = 0$$

$$\Rightarrow A \Sigma P \Lambda P^T P = 0 \cdot P = 0$$

$$\Rightarrow A \Sigma P \Lambda = 0$$

we know ^{that} any function of BX , $f(BX)$, is also mutually independent of AX .

$$\text{Let } f(x) = \prod_{i=1}^n \frac{1}{\lambda_i} x_i^2$$

$$BX = P \Lambda P^T X = P \Lambda Z \quad P^T X = [p_1 x_1 \dots p_n x_n]^T = [z_1 \dots z_n]^T$$

$$f(BX) = \prod_{i=1}^n \frac{1}{\lambda_i} (p_i \lambda_i z_i)^2 = \prod_{i=1}^n (p_i \lambda_i^{1/2} z_i)^2 = [P \Lambda^{1/2} Z]^T [P \Lambda^{1/2} Z]$$

$$= Z^T \Lambda^{1/2} P^T P \Lambda^{1/2} Z$$

$$= Z^T \Lambda Z$$

$$= (P^T X)^T \Lambda (P^T X) = X^T P \Lambda P^T X = X^T B X$$

$$(\Leftarrow) AX \text{ and } X^T B X \text{ are independent.} \Rightarrow A \Sigma B = 0$$

$$\text{From Theorem 11, } \text{Cov}(X, X^T B X) = 2 \Sigma B \mu$$

$$\therefore \text{Cov}(AX, X^T B X) = 2 A \Sigma B \mu = 0$$

$$\therefore A \Sigma B = 0 \quad \square$$

(b)

Q10) Solve Example 12

$$X_1, \dots, X_n \sim N_n(\mu, \sigma^2)$$

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \quad \text{and} \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

show that $\bar{X}$ and S^2 are independent

$$\bar{X} = \Pi_{\underline{1}} X \quad \text{and} \quad S^2 = \frac{1}{n-1} X^T (I_n - \Pi_{\underline{1}}) X$$

$$\text{Let } A = \Pi_{\underline{1}} \quad \text{and} \quad B = (I_n - \Pi_{\underline{1}})$$

$$ASB = \Pi_{\underline{1}} \cdot \sigma^2 I_n \cdot (I_n - \Pi_{\underline{1}}) = \sigma^2 \Pi_{\underline{1}} (I_n - \Pi_{\underline{1}}) = 0 \quad \therefore \bar{X} \text{ and } S^2 \text{ are independent}$$

Q11) Prove Theorem 13

(a) A is idempotent $\Leftrightarrow Z^T A Z \sim \chi^2(r)$, $r = \text{rank}(A) = \text{tr}(A)$

(b) $A\Sigma$ is idempotent $\Leftrightarrow X^T A X \sim \chi^2(r)$, $r = \text{rank}(A)$

$$Z \sim N_n(0, I_n), \quad A^T = A, \quad X \sim N_n(0, \Sigma), \quad \Sigma: p.d.$$

(a) $(\Rightarrow)$

$$A = P \Lambda P^T, \quad P P^T = P^T P = I_n, \quad \Lambda = \text{diag}(\lambda_1) \text{ is } n$$

$$Z^T P \Lambda P^T Z = (P^T Z)^T \Lambda (P^T Z) = Z^T \Lambda Z \quad \therefore P^T Z \sim N(P^T 0, P^T P) \stackrel{d}{=} N(0, I_n)$$

$$= \sum_{i=1}^n \lambda_i Z_i^2 = \sum_{i=1}^r Z_i^2 \quad \therefore \Lambda = \text{diag}(\underbrace{1, \dots, 1}_{r \text{ times}}, \underbrace{0, \dots, 0}_{n-r \text{ times}}), \quad r = \text{rank}(A)$$

$$Z_i^2 \sim \chi^2(1) \quad \therefore \sum_{i=1}^r Z_i^2 \sim \chi^2(r)$$

($\Leftarrow$) $Z^T A Z \sim \chi^2(r)$

$$\begin{aligned} \text{mgf of } Z^T A Z: M_{Z^T A Z}(t) &= E\left(e^{t \sum_{i=1}^r \lambda_i Z_i^2}\right) = E\left[e^{t \lambda_1 Z_1^2} \dots e^{t \lambda_r Z_r^2}\right] \\ &= (1 - 2t\lambda_1)^{-1/2} \dots (1 - 2t\lambda_r)^{-1/2} \\ &= (1 - 2t)^{-r/2} \quad \therefore \lambda_1 = \dots = \lambda_r = 1 \end{aligned}$$

$\Leftrightarrow$ eigenvalue of A are either 0 or 1

$\Leftrightarrow$ A is idempotent

(b) $X^T A X = (\Sigma^{1/2} Z)^T A (\Sigma^{1/2} Z) = Z^T \Sigma^{1/2} A \Sigma^{1/2} Z$

if $A\Sigma$ is idempotent $A\Sigma = A\Sigma A\Sigma \Rightarrow \Sigma^{1/2} A \Sigma^{1/2} = \Sigma^{1/2} A \Sigma A \Sigma^{1/2}$

$\therefore \Sigma^{1/2} A \Sigma^{1/2}$ is idempotent

$\therefore X^T A X \sim \chi^2(r)$, $r = \text{rank}(\Sigma^{1/2} A \Sigma^{1/2}) = \text{rank}(A) \quad \therefore \Sigma^{1/2}$ is non-singular & invertible matrix exists.

to show reverse argument, go backwards & use (a)

Q12) Prove Theorem 15

The following statements are equivalent:

(a) $Y \sim \chi^2(K, \delta)$

(b) $M_Y(t) \sim (1-2t)^{-K/2} \exp\left\{-\frac{\delta t}{1-2t}\right\}, \quad t < 1/2$

$\vdots$

(d) $Y \stackrel{d}{=} V + Z^2, \quad V \sim \chi^2(K-1), \quad Z \sim N(0, 1), \quad \theta = \delta$

$$\text{Hint: } Y \stackrel{d}{=} \sum_{i=1}^K Z_i^2, \quad \delta = \sum_{i=1}^K \mu_i^2, \quad Z_i \sim N(\mu_i, 1)$$

* a $\Leftrightarrow$ b : derive $M_{Z_i^2}(t)$ first. Don't forget the range of t .

(a) $\Leftrightarrow$ (b)

$$Z \sim N(\mu, 1), \quad M_{Z^2}(t) = E[e^{tZ^2}] = \int_{-\infty}^{+\infty} e^{tZ^2} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}(Z-\mu)^2\right] dz$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \exp\left\{tZ^2 - \frac{1}{2}(Z-\mu)^2\right\} dz$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \exp\left\{-\frac{(1-2t)Z^2 - 2\mu Z + \mu^2}{2}\right\} dz$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \exp\left\{-\frac{(1-2t)}{2} \left(Z^2 - \frac{2\mu Z}{(1-2t)} + \left(\frac{\mu}{1-2t}\right)^2\right) + \frac{\mu^2}{1-2t}\right\} dz$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \exp\left\{-\frac{(1-2t)}{2} \left(Z - \frac{\mu}{1-2t}\right)^2 + \frac{\mu^2}{2(1-2t)} - \frac{\mu^2}{2}\right\} dz$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \exp\left\{-\frac{1}{2(1-2t)} \left(Z - \frac{\mu}{1-2t}\right)^2\right\} \cdot \exp\left\{\frac{2t\mu^2}{2(1-2t)}\right\} dz$$

$$= \exp\left\{\frac{t\mu^2}{1-2t}\right\} (1-2t)^{-1/2} \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}(1-2t)} \exp\left\{-\frac{1}{2} \frac{1}{(1-2t)} \left(Z - \frac{\mu}{1-2t}\right)^2\right\} dz$$

$$= (1-2t)^{-1/2} \exp\left\{\frac{t\mu^2}{1-2t}\right\}$$

$$\begin{aligned} M_Y(t) &= M_{Z_1^2}(t) \dots M_{Z_K^2}(t) = (1-2t)^{-K/2} \exp\left\{\frac{t}{1-2t} (\mu_1^2 + \dots + \mu_K^2)\right\} \\ &= (1-2t)^{-K/2} \exp\left\{\frac{t}{1-2t} \cdot \delta\right\}, \quad \delta = \mu_1^2 + \dots + \mu_K^2 \end{aligned}$$

(b) $\Leftrightarrow$ (d)

$Y \sim \chi^2(K, \delta) \Leftrightarrow Y = \sum_{i=1}^K Z_i^2$

$Z_i \sim N(\mu_i, 1)$, Z_i ($i=1, \dots, K$) are mutually independent and $\delta = \sum_{i=1}^K \mu_i^2$

let $\mu_1 = \dots = \mu_{K-1} = 0$ and $\mu_K = \sqrt{\delta}$

$$Y = Z_1^2 + \dots + Z_{K-1}^2 + Z_K^2$$

$$Z_i \stackrel{iid}{\sim} N(0, 1) \quad (i=1, \dots, K-1)$$

$$\therefore \sum_{i=1}^{K-1} Z_i^2 \stackrel{d}{=} V \sim \chi^2(K-1) \quad \text{and} \quad Z_K = Z \sim N(\sqrt{\delta}, 1)$$

V and Z are independent

Thus V and Z^2 are independent.

Q13) Prove Theorem 13

(a) $Z \sim N_n(\mu, I_n)$, A is real and symmetric

A is idempotent $\Leftrightarrow Z^T A Z \sim \chi^2(r; \delta)$, $r = \text{rank}(A)$, $\delta = \mu^T A \mu$

(b) $X \sim N_n(\mu, \Sigma)$, Σ is positive definite

$A \Sigma$ is idempotent $\Leftrightarrow X^T A X \sim \chi^2(r; \delta)$, $r = \text{rank}(A)$, $\delta = \mu^T A \mu$

(a) $(\Leftarrow) Z \sim N_n(\mu, I_n)$, $A = A^T \Rightarrow A = P \Lambda P^T$, $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_r, 0, \dots, 0)$

$$P P^T = P^T P = I_n$$

$$Z^T A Z = Z^T (P \Lambda P^T) Z = (P^T Z)^T \Lambda (P^T Z)$$

$$P^T Z \sim N(P^T \mu, P^T P) \stackrel{d}{=} N(P^T \mu, I_n)$$

$$\text{Let } W = P^T Z, \quad Z^T A Z = W^T \Lambda W$$

$$\text{Let } \xi = P^T \mu, \quad \xi = [\xi_1, \dots, \xi_n]^T$$

$\Rightarrow W_i \sim N(\xi_i, 1)$, W_i and W_j are mutually independent ($i \neq j$)

$$\therefore \text{Cov}(W_i, W_j) = 0$$

$$M_{Z^T A Z}(t) = M_{W^T \Lambda W}(t) = E\left[\exp\left\{t \cdot \sum_{i=1}^n \lambda_i W_i^2\right\}\right]$$

$$= \prod_{i=1}^n E\left[\exp\{t \cdot \lambda_i W_i^2\}\right] = \prod_{i=1}^n E\left[\exp\{t \cdot \lambda_i W_i^2\}\right] \quad (\because \lambda_{r+1} = \dots = \lambda_n = 0)$$

$$= \prod_{i=1}^r M_{\lambda_i W_i^2}(t) = \prod_{i=1}^r (1 - 2t)^{-1/2} \exp\left\{\frac{\xi_i^2 \lambda_i t}{1 - 2t}\right\} \quad (\because W_i \sim \chi^2(1; \xi_i^2))$$

$$\text{Because } Z^T A Z = W^T \Lambda W \sim \chi^2(r; \delta), \quad M_{W^T \Lambda W}(t) = (1 - 2t)^{-1/2} \exp\left\{\frac{\delta t}{1 - 2t}\right\}, \quad t < 1/2$$

$$\therefore \lambda_1 = \dots = \lambda_r = 1 \quad \text{and} \quad \sum_{i=1}^n \lambda_i \xi_i^2 = \delta$$

Note that $\sum_{i=1}^n \lambda_i \xi_i^2 = \xi^T \Lambda \xi = (P^T \mu)^T \Lambda (P^T \mu) = \mu^T P \Lambda P^T \mu = \mu^T A \mu$

$\therefore \mu^T A \mu = \delta$ and A is idempotent ($\because \lambda_i = 0$ or 1)

$$(\Rightarrow) Z^T A Z = Z^T P \Lambda P^T Z = (P^T Z)^T \Lambda (P^T Z)$$

$$= W^T \Lambda W \quad | \quad W = P^T Z, \quad W \sim N(P^T \mu, I_n) \stackrel{d}{=} N(\xi, I_n)$$

$$= \sum_{i=1}^r \lambda_i W_i^2$$

$$\because A \text{ is idempotent, } \Lambda = \underbrace{\text{diag}(1, \dots, 1)}_{r \times r} \underbrace{\text{diag}(0, \dots, 0)}_{n-r \times n-r}$$

$$\therefore Z^T A Z = \sum_{i=1}^r W_i^2 \sim \chi^2(r, \sum_{i=1}^r \xi_i^2) \stackrel{d}{=} \chi^2(r, \sum_{i=1}^n \lambda_i \xi_i^2) \stackrel{d}{=} \chi^2(r, \mu^T A \mu)$$

Let $\delta = \mu^T A \mu$. Then $Z^T A Z \sim \chi^2(r, \delta)$

(b) Let $Z = \Sigma^{-1/2} X \sim N_n(\Sigma^{-1/2} \mu, I_n)$

$$\delta = (\Sigma^{-1/2} \mu)^T \Sigma^{1/2} A \Sigma^{1/2} (\Sigma^{-1/2} \mu) = \mu^T A \mu$$

$$X^T A X = (\Sigma^{1/2} Z)^T A (\Sigma^{1/2} Z) = Z^T \Sigma^{1/2} A \Sigma^{1/2} Z \sim \chi^2(r; \delta), \quad r = \text{rank}(\Sigma^{1/2} A \Sigma^{1/2})$$

iff $\Sigma^{1/2} A \Sigma^{1/2}$ is idempotent

* $\text{rank}(\Sigma^{1/2} A \Sigma^{1/2}) = \text{rank}(A) \because \Sigma^{1/2}$ is invertible / full rank

$$+ \Sigma^{1/2} A \Sigma^{1/2} = \Sigma^{1/2} A \Sigma A \Sigma^{1/2}$$

$$\Rightarrow A \Sigma^{1/2} = A \Sigma A \Sigma^{1/2}$$

$$\Rightarrow A \Sigma = A \Sigma A \quad \therefore A \Sigma \text{ is idempotent}$$

$\therefore X^T A X \sim \chi^2(r; \delta), \delta = \mu^T A \mu, r = \text{rank}(A)$ iff $A \Sigma$ is idempotent

Q14) Definition of non-central chi-Square distribution

$$Y \sim \chi^2(K; \delta), \quad Y \stackrel{d}{=} \sum_{i=1}^K Z_i^2, \quad Z_i \sim N(\mu_i, 1)$$

$$\delta = \sum_{i=1}^n \mu_i^2$$

Definition of non-central F-distribution

$$Y \sim F(K_1, K_2; \delta), \quad Y \stackrel{d}{=} \frac{V_1 / K_1}{V_2 / K_2}, \quad V_1 \sim \chi^2(K_1; \delta) \text{ and } V_2 \sim \chi^2(K_2)$$

* Remark

$$\text{if } T \sim t(K; \delta), \quad T^2 \sim F(1, K; \delta^2)$$

$$\therefore T = \frac{Z}{\sqrt{V/K}}, \quad T^2 = \frac{Z^2/1}{V/K}, \quad Z \sim N(\delta; 1) \text{ and } V \sim \chi^2(K)$$

Z and V are independent.

$$Z^2 \sim \chi^2(1; \delta^2)$$

$$\therefore T^2 = \frac{V_1/1}{V_2/K}, \quad V_1 \sim \chi^2(1; \delta^2) \text{ and } V_2 \sim \chi^2(K)$$

$$\sim F(1, K; \delta^2)$$

Q15) Prove Example 17

$$X_1, \dots, X_n \sim N(\mu, \sigma^2)$$

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \text{ and } S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

$$\text{Show that } \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t(n-1; \delta), \quad \delta = \frac{\mu - \mu_0}{\sigma/\sqrt{n}}$$

$$\frac{\bar{X} - \mu_0}{S/\sqrt{n}} = \frac{\bar{X} - \mu + \mu - \mu_0}{\sigma/\sqrt{n}} = \frac{\sqrt{\frac{(n-1)S^2}{\sigma^2}}}{1} \bigg/ \sqrt{\frac{(n-1)}{\sigma^2}}$$

$$= \frac{Z + \delta}{\sqrt{V/(n-1)}}, \quad Z \sim N(0, 1), \quad V \sim \chi^2(n-1)$$

$$\therefore \frac{Z + \delta}{\sqrt{V/(n-1)}} \sim t(n-1; \delta), \quad \delta = \frac{\mu - \mu_0}{\sigma/\sqrt{n}}$$